

PH 612: Nuclear and Particle Physics

Course Code	PH 612
Course Name	Nuclear and Particle Physics
L-T-P-C	3-0-0-3
Prerequisites	Quantum Mechanics, Electromagnetic Theory
Intended for	Core for M.Sc. and I-Ph.D. in Physics; Elective for B.Tech. 3rd and 4th Years.
Mutual Exclusion	NA
Approval	10th Senate (Jun 2016)
Revised	64th BoA (May 2026) - syllabus revised

Preamble

The objective of the proposed course is to introduce students to the fundamental principles and concepts of nuclear and particle physics. Students will be able to know the fundamentals of the interaction of high-energy particles. This course is expected to provide a working knowledge of real-life problems.

Course Modules

- **Properties of Nuclei:** Nuclear size, nuclear radius and charge distribution, Rutherford Scattering, mass and binding energy, semi-empirical mass formula, nuclear stability, angular momentum, parity and isospin, magnetic dipole moment, electric quadrupole moment, and nuclear shape. (5 Lectures)
- **Nuclear models:** Evidence of shell structure (magic numbers), Liquid drop model, Shell model, ground state spin & parity. (4 Lectures)
- **Two-body problems:** Deuteron ground state, excited states, spin dependence of nuclear forces, two-nucleon scattering, charge symmetry and charge independence of nuclear forces, exchange nature of nuclear forces. (4 Lectures)
- **Nuclear decay:** Particle decays, Gamow theory of Alpha decay, Fermi theory of Beta decay, Gamma decay, direct evidence for the neutrino. (5 Lectures)
- **Nuclear Reactions:** Different types of reactions, Direct & Compound nucleus formation and break-up, nuclear fission, fusion reaction. (4 Lectures)
- **Elementary particles:** Fundamental interactions. Particle Zoo: Leptons, Hadrons. Organizing principle: Baryon and Lepton Numbers, Strangeness, Isospin, The eightfold way. Quarks: Colour charge and strong interactions, confinement, Gell-Mann–Nishijima formula, Relativistic kinematics and conservation laws. The Standard Model: parity violation in weak interaction, Wu's experiment, elementary idea about electroweak unification, quark model, concept of colour charge, discrete symmetries, properties of quarks and leptons, particle interactions, and Feynman diagrams. (15 Lectures)
- **Experimental Techniques and Applications:** Particle Accelerators: LINAC, Cyclotron, Synchrotron, Brief overview on present-day accelerators (RHIC, LHC). Particle detectors: Geiger-Muller counter, scintillators, calorimeters, and trackers. Radiation therapy, nuclear medicine, and imaging, dosimetry. (5 Lectures)

Text Books

1. Kenneth S Krane, *Introductory Nuclear Physics*, Vol., Wiley, UK, 2016
2. David Griffiths, *Introduction to Elementary Particles* (4th Edition)

References

1. *An introduction to the physics of Nuclei and Particles* by Richard A Dunlap (2004)
2. *Nuclear and Particle Physics: An Introduction* by B. R. Martin (2013)
3. *Nuclear Physics: Principles and Applications* by John Lilley (2001)
4. *Modern Particle Physics* by Mark Thomson (2013)